

VORTEX-INDUCED VIBRATIONS IN IDLING WIND TURBINE BLADES: INFLUENCE OF WIND SPEED AND PITCH ANGLE WITH MITIGATION RECOMMENDATIONS

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The vortex-induced vibration (VIV) of idling wind turbine blades of 15MW IEA reference offshore wind turbine is studied. During idling, the blades are pitched out (feathered) to 95 degrees. VIV of blades is often observed, consuming the fatigue life of the blade. The frequency of alternate vortex-shedding, given by the Strouhal number, depends on the wind speed and the angle of attack, which varies along the blade span. The blade is modelled as a twisted tapered, hollow cantilever beam. The coupled flapwise-edgewise-torsional free vibration is analysed through the Rayleigh-Ritz method to generate the natural frequencies, and check for the tuning factors at various wind speeds and pitching angles. The oscillating components of the aerodynamic force coefficients are generated through 2D CFD, using dynamic meshing and RANS $k - \omega$ SST model. This leads to the VIV-excitation forces along the blade span. This is followed by the coupled dynamic response, generated through the modal superposition principle, for distinct wind speeds and pitch angles. An efficient machine learning model has been used to predict blade root stresses, trained on a dataset comprising wind speed, pitch angle, and corresponding oscillatory aerodynamic coefficients. Through this sensitivity study, the optimal pitch angle at each wind speed is recommended.

Keywords: Blade, Vortex-Induced Vibration, Dynamic meshing, Modal superposition, Gaussian Process Regressor.

1. Introduction

Wind energy contributes >10% of world's renewable energy. As wind turbines become more powerful, the blade size increases to capture more energy. Longer blades are more flexible, and hence, susceptible to dynamic loads like wind turbulence, rotating unbalance, and vortex-shedding. Dynamic loads cause reversing stresses in blades, in both flap-wise and edge-wise modes, consuming the fatigue life. A turbine spends about 98% of its life in production, and only 2% in idling (low wind speeds, extreme weather, maintenance/repair). However, idling accounts for 5-20% of the accumulated fatigue damage. Blade vibration during idling may be stall-induced (non-periodic, broadband), vortex-induced (periodic, narrow-band), or due to turbulence (stochastic, broadband).

Vortex-induced vibration (VIV) of blades has been a subject of recent research. Chen et al. (2025) presented a nonlinear model of VIV in wind turbine towers under parked conditions, showing that VIV can be mitigated by changing azimuth angles and pitch angles. Horcas et al. (2022) investigated the 10MW IEA Wind turbine to check whether trailing edge flaps can mitigate VIV in large wind turbine blades during standstill conditions. Pirrung et al. (2024) extended VIV studies from single-blade analyses to full rotor simulations of the IEA 10 MW reference turbine, implying that rotor-level coupling amplifies VIV risk significantly. Loubeyres et al. (2022) investigated stall flutter instabilities

in the IEA-15MW reference wind turbine under idling conditions with strong crosswinds, showing the IEA-15MW design is particularly susceptible to stall flutter in idling conditions. Horcas et al. (2020) analysed how blade tip geometry influences VIV in large wind turbine blades, showing that tip sweep emerges as a promising passive mitigation strategy. Bidadi et al. (2024) presented the first 3D computational study of wind turbine aerofoils over the full 360° angle-of-attack range at Reynolds' number $Re = 10^7$ to assess unsteady aerodynamic loads and vortex-shedding characteristics, showing that deep stall regimes can trigger VIV at large angles of attack. Ma et al. (2025) studied the aeroelastic performance of the IEA-15MW wind turbine under yaw conditions. Saram and Yang (2024) analysed coupled flap-wise and edge-wise modes in the 5MW NREL HAWT blades, focusing on nonlinear interactions, showing that coupled-mode analysis is essential for predicting blade dynamics in large turbines. Heinz et al. (2014) investigated VIV in the DTU 10-MW reference wind turbine blade under standstill conditions, showing inclined inflow with velocity components along the blade axis can trigger spanwise correlated vortex-shedding, confirming that standstill VIV is a realistic risk for modern large blades. Grinderslev et al. (2023) examined forced-motion CFD simulations of VIV in wind turbine blades, focusing on sensitivity to turbulence models, grids, and solver choices.

The **aim** of this work is to investigate the vortex-induced vibration of wind turbine blades in idling conditions using the IEA 15MW reference offshore wind turbine, at various wind speeds and pitch angles. The **objective** is to minimize the blade root stress due to VIV for various pitch angles, for a wide range of wind speeds, which can serve as operational recommendations. The **layout** of the paper is as follows: (a) The vortex-shedding frequency is quantified for various wind speeds and pitch angles along the blade span; (b) The free vibration of the blade is analysed for the natural frequencies, and its resonance with the vortex-shedding frequencies are checked; (c) 2D CFD analysis is done for the different aerofoils to generate the oscillating aerodynamic coefficients (d) Coupled dynamic response analysis using modal superposition is performed to generate the root stress, and (e) Optimum pitch angle for various wind speeds is recommended, supported by Machine Learning.

2. Problem formulation

2.1 Geometrical parameters

The IEA 15MW reference offshore wind turbine blade (**Figure 1**) is considered for analysis of vortex-induced vibration. The primary parameters for the blade vibration analysis are given in **Table 1**. The blade is composed of seven different aerofoils from FFA-W3 series, each running over a part (sub-span) of the blade. The span-wise distributions of chord $c(x)$, thickness, and twist $\beta(x)$, the span-wise distributions of geometric and structural properties (mass, stiffness) are known from [1]. The blade is tapered from the root $x = 0$ to the tip $x = L$. The sections closer to the root provide structural strength, while the sections closer to the tip generate most of the aerodynamic forces. The angle between the chord-line (nose-tail line) and the rotor-plane is the twist angle $\beta(x)$. This is designed to optimize the angle of attack (AoA) $\alpha(x)$ for most of the blade span. Structural damping is neglected; however, aerodynamic damping effects are implicitly captured through URANS-based force coefficients. In the idling condition (**Figure 2**), the inflow angle of the wind is 90° . The angle of attack $\alpha(x) = 90^\circ - \beta(x)$. Let the controller cause the blade to pitch by an angle δ , measured anti-clockwise from the rotor plane (**Figure 3**). Due to the twist $\beta(x)$, each section has a different $\alpha(x)$ for the same δ . Thus, $\alpha(x) = 90^\circ - \beta(x) - \delta$. For computation efficiency, each section aerofoil is individually considered in CFD, with unit chord, various δ , and the mean AoA $\bar{\alpha}$ over its sub-span (**Table 2**).

Table 1: Primary parameters for IEA 15MW offshore Wind Turbine blade vibration analysis.

Description	Magnitude	Unit	Description	Magnitude	Unit
Blade length L	117	m	Blade mass	65,250	kg
Fluid density ρ_a	1.225	kg/m^3	First flap-wise frequency	0.555	Hz
Fluid viscosity $\nu _{20^\circ C}$	1.51×10^{-5}	$m^2.s$	First edge-wise frequency	0.642	Hz

Table 2 : Mean AoA ($\bar{\alpha}$) at various pitch angles δ .

				$\delta = 105^\circ$	$\delta = 95^\circ$	$\delta = 85^\circ$	$\delta = 75^\circ$	$\delta = 60^\circ$
#	Blade Section	$\bar{c}(x)$	$x \downarrow$	$\bar{\alpha}$	$\bar{\alpha}$	$\bar{\alpha}$	$\bar{\alpha}$	$\bar{\alpha}$
1	SNL-FFA-W3-500	5.73	19.10	-23.67	-13.67	-3.67	6.33	21.33
2	FFA-W3-360	5.39	31.04	-20.66	-10.66	-0.66	9.34	24.34
3	FFA-W3-330blend	4.77	40.59	-18.48	-8.48	1.52	11.52	26.52
4	FFA-W3-301	4.21	52.53	-16.84	-6.84	3.16	13.16	28.16
5	FFA-W3-270blend	3.71	64.46	-15.64	-5.64	4.36	14.36	29.36
6	FFA-W3-241	3.20	76.40	-14.41	-4.41	5.59	15.59	30.59
7	FFA-W3-211	2.22	90.73	-13.22	-3.22	6.78	16.78	31.78

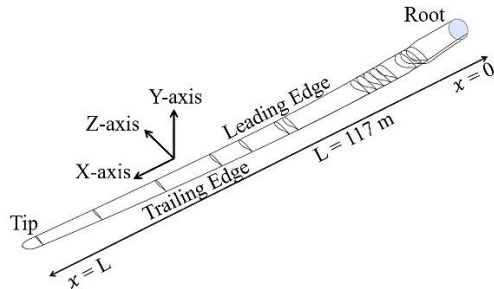


Figure 1 : IEA 15MW OWT Blade CAD

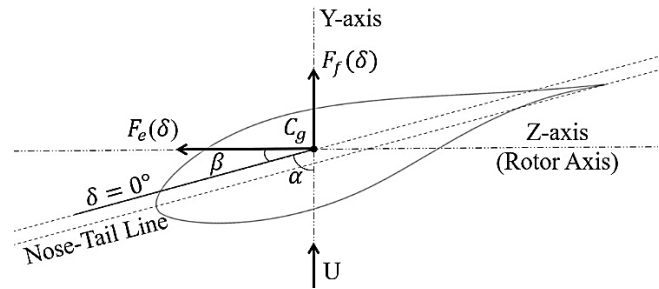
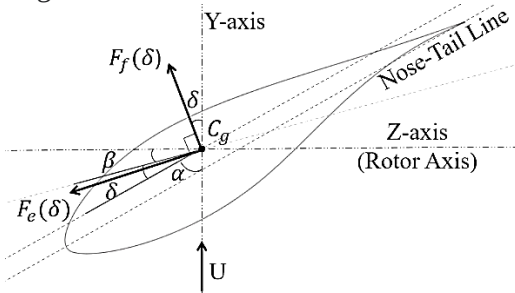
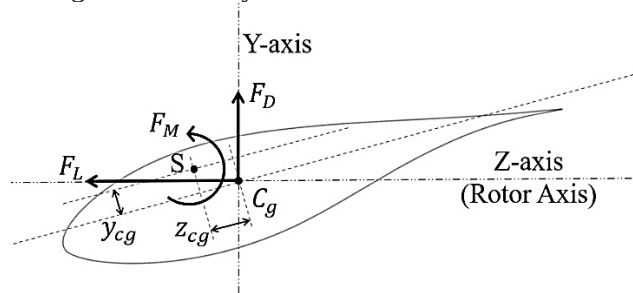

 Figure 2 : Aerodynamic Forces at zero $\delta = 0$.

 Figure 3 : Aerodynamics forces at pitching angle δ .


Figure 4 : Coupled vibration excitation forces.

2.2 Vortex-shedding: Strouhal number

Flow past bluff bodies is associated with shedding of vortices downstream. At certain Reynolds numbers Re , this vortex-shedding alternates between the left and right of the body. The alternating low-pressure zones lead to cross-flow structural oscillations. Vortex-shedding over the blade becomes prominent at idling, with $Re(x) = Uc(x)/\nu|_{20^\circ\text{C}} \sim 10^6$, for wind speed $1 \leq U \leq 15 \text{ m/s}$. The frequency of alternate vortex-shedding is given by the Strouhal number $St|_{Re=10^6}(\alpha) = f_{viv}(x; \alpha)c(x)/U$. **Table 3** shows the flow regimes and wake patterns for FFA-W3 series aerofoils and associated St , for various AoA α . At very low α , when the flow remains attached to the aerofoil, the vortex-shedding frequencies are higher due to shear layer instabilities inside the boundary layer. Lift is strong, drag is negligible. At around the stall α , the flow begins to detach from the aerofoil, the wake becomes wider and turbulent, and the shedding frequency drops. Lift begins to drop sharply, and drag rises. At large α , the aerofoil behaves like a bluff body. There is strong flow separation and a broad turbulent wake. The vortex-shedding frequency drops further. Lift is negligible, drag is dominant. Thus, $f_{viv}(x; \alpha) = St(\alpha) \cdot U/c(x)$ varies along the span; causing a non-harmonic excitation. If f_{viv} nears one of the blade natural frequencies f , there are resonant vibrations and large dynamic stresses.

Table 3 : Strouhal number vs. AoA for FFA-W3 series aerofoils of 15MW IEA reference OWT Blade.

AoA	$\alpha < -19^\circ, \alpha > 17^\circ$	$-19^\circ < \alpha < -5^\circ, 12^\circ < \alpha < 17^\circ$	$-5^\circ < \alpha < 12^\circ$
Flow regime	Post-Stall	Near Stall	Attached flow
Vortex-shedding	Large-scale	Partial flow separation	Strong bound circulation
$St(\alpha)$	0.2	0.3	0.4

The aerodynamic forces need to be resolved in the blade's edge-wise and flap-wise directions using the pitch angle δ . The aerodynamic force along the direction of the wind is Drag F_D , causing flap-wise vibration; the aerodynamic force perpendicular to the wind is the Lift F_L , causing edge-wise vibration, and the aerodynamic moment F_M causes the torsional vibration; formulated as:

$$F_D(x, t; \alpha) = \frac{1}{2} \rho_a c(x) U^2 \mu(x) [C_D(\alpha, x) + \Delta C_D(\alpha, x) \sin(2\pi f_{viv}(x; \alpha)t)] \dots (N/m) \quad (1a)$$

$$F_L(x, t; \alpha) = \frac{1}{2} \rho_a c(x) U^2 \mu(x) [C_L(\alpha, x) + \Delta C_L(\alpha, x) \sin(2\pi f_{viv}(x; \alpha)t)] \dots (N/m) \quad (1b)$$

$$F_M(x, t; \alpha) = \frac{1}{2} \rho_a \{c(x)\}^2 U^2 \mu(x) [C_M(\alpha, x) + \Delta C_M(\alpha, x) \sin(2\pi f_{viv}(x; \alpha)t)] \dots (Nm/m) \quad (1c)$$

Since the blade is a 3D body, the Prandtl's tip/root-loss correction factors $\mu(x) = \mu_{tip}(x)\mu_{root}(x)$ are included. (Burton et al (2001)). The use of 2D sectional CFD neglects spanwise flow correlation, which will be addressed in future 3D studies. In Eq(1), the static component causes a static deflection, while the oscillating component causes the dynamic excitation force at the vortex-shedding frequency $f_{viv}(x; \alpha)$. The static coefficients $C_D(\alpha, x)$, $C_L(\alpha, x)$, $C_M(\alpha, x)$ are generated when the aerofoil is at rest, and are available in [1]. The dynamic coefficients $\Delta C_D(\alpha, x)$, $\Delta C_L(\alpha, x)$, $\Delta C_M(\alpha, x)$, are generated using 2D CFD, with the aerofoil moving. All coefficients are functions of Re and α . Since the coupled blade vibration is analysed w.r.t the global coordinate system, pitching the blade modifies the edge-wise and flap-wise excitation forces in Eq(1) through coordinate-transformation as:

$$F_f(x, t; \delta) = F_D(x, t; \alpha) \cos \delta + F_L(x, t; \alpha) \sin \delta; \quad F_e(x, t; \delta) = F_L(x, t; \alpha) \cos \delta - F_D(x, t; \alpha) \sin \delta \quad (2)$$

3. Vortex-induced vibration

The coupled system of governing differential equations (GDE) for flap-wise and edge-wise flexural vibration (balancing forces about the CoG of the aerofoil (C_g)) and torsional vibration (by balancing moments about the shear centre (S)), as shown in **Figure 4**, is

$$m(x)\ddot{u} + m(x)y_{cg}(x)\ddot{\theta} + EI_{zz}(x)u'''' + EI_{yz}(x)w'''' + EI_{zz}K(x)\theta'' = F_f(x, t) \quad (3a)$$

$$m(x)\ddot{w} + m(x)z_{cg}(x)\ddot{\theta} + EI_{yz}(x)u'''' + EI_{yy}(x)w'''' + EI_{yy}K(x)\theta'' = F_e(x, t) \quad (3b)$$

$$m(x)y_{cg}(x)\ddot{u} + m(x)z_{cg}(x)\ddot{w} + \rho I_p(x)\ddot{\theta} + EI_{zz}K(x)u'' + EI_{yy}K(x)w'' + GJ(x)\theta'' = F_M(x, t) \quad (3c)$$

where $u(x, t)$ is the flap-wise displacement, $w(x, t)$ is the edge-wise displacement, and $\theta(x, t)$ is the angular displacement, with the other symbols having their usual meaning in beam vibration. From [1], the inertia and stiffness terms are known over the span. Writing in the matrix form, Eq(3) is

$$\begin{bmatrix} m(x) & 0 & m(x)y_{cg} \\ 0 & m(x) & m(x)z_{cg} \\ m(x)y_{cg} & m(x)z_{cg} & \rho I_p(x) \end{bmatrix} \begin{Bmatrix} u(x, t) \\ w(x, t) \\ \theta(x, t) \end{Bmatrix} + \begin{bmatrix} EI_{zz}(x) & EI_{yz}(x) & EI_{zz}K(x) \\ EI_{yz}(x) & EI_{yy}(x) & EI_{yy}K(x) \\ EI_{zz}K(x) & EI_{yy}K(x) & GJ(x) \end{bmatrix} \begin{Bmatrix} u(x, t) \\ w(x, t) \\ \theta(x, t) \end{Bmatrix} = \begin{Bmatrix} F_D(x, t) \\ F_L(x, t) \\ F_M(x, t) \end{Bmatrix} \quad (4)$$

which can be written more compactly as $[M]\{\ddot{X}(t)\} + [K]\{X(t)\} = \{F(t)\}$.

3.1 Free Vibration

The homogenous set of equations $[M]\{\ddot{X}(t)\} + [K]\{X(t)\} = \{0\}$ represent the coupled free vibration, which is solved by the Rayleigh-Ritz (R-R) method. The admissible functions for the flap-wise and edge-wise bending vibration are the first two (02) modes $\phi_u(x)$, $\phi_w(x)$ of a uniform cantilever beam, while those of torsional vibration are the first two (02) modes $\phi_\theta(x)$ of a uniform cantilever. According to R-R formulation, the blade modeshapes $\Phi_j(x)$ is assumed to be weighted summation of the admissible functions, i.e., $\Phi_j(x) = \sum_{k=1}^N C_{jk} \phi_k(x)$, where $\phi_k(x)$ represents the k^{th} admissible function with coefficient C_k . The R-R method formulates the natural frequency ω in terms of maximum potential and kinetic energies as $\omega^2 = \frac{U_{max}}{T_{max}}$, where T_{max} and U_{max} are expressed in terms of generalized mass and generalized stiffness as : $U_{max} = (1/2) \sum_i \sum_j G K_{ij} C_i C_j$, $T_{max} = (1/2) \sum_i \sum_j G M_{ij} C_i C_j$. The generalized mass and stiffness matrices are : $G M_{11} = \int_0^L m(x) \phi_u^2(x) dx$;

$GM_{22} = \int_0^L m(x)\phi_w^2(x)dx$; $GM_{66} = \int_0^L I_p(x)\phi_\theta^2(x)dx$; $GM_{12} = GM_{21} = 0$; $GM_{26} = GM_{62} = \int_0^L m(x)z_{cg}(x)\phi_w(x)\phi_\theta(x)dx$; $GM_{16} = GM_{61} = \int_0^L m(x)y_{cg}(x)\phi_u(x)\phi_\theta(x)dx$; $GK_{11} = \int_0^L EI_{zz}(x)[\phi_u''(x)]^2dx$; $GK_{22} = \int_0^L EI_{yy}(x)[\phi_w''(x)]^2dx$; $GK_{66} = \int_0^L GJ(x)[\phi_\theta'(x)]^2dx$; $GK_{16} = GK_{61} = \int_0^L EI_{zz}(x)K(x)\phi_u''(x)\phi_\theta'(x)dx$; $GK_{26} = GK_{62} = \int_0^L EI_{yy}(x)K(x)\phi_w''(x)\phi_\theta'(x)dx$; $GK_{12} = GK_{21} = \int_0^L EI_{yz}(x)\phi_u''(x)\phi_w''(x)dx$. Minimizing the frequency with respect to C_k , the system of equations is solved to compute the weights for admissible functions and the natural frequencies $\omega = 2\pi f$. The non-trivial solution generates N distinct sets of the coefficients, i.e., $\{C_1, C_3, C_5, \dots, C_N\}$, along with N natural frequencies of the blade. Once the coefficients C_k are known, $\Phi_j(x)$ is generated.

3.2 Dynamic Response: Normal Mode Summation

Using the principle of modal superposition of modeshapes and principal coordinates $\{q(t)\}$, i.e., the system of generalized (modal) GDE becomes $[GM]\{\ddot{q}(t)\} + [GK]\{q(t)\} = \{GF(t)\}$. (5) The displacement vector is given as $\{X(x, t)\}^T = \{u(x, t) \ v(x, t) \ w(x, t)\}^T = [\phi(x)]\{q(t)\}$; and the generalized forcing is given as $\{GF(t)\}^T = \{G_f(t) \ G_e(t) \ G_M(t)\}$; i.e., $G_f(t) = \int_0^L \phi(x)F_f(x, t)dx$; $G_e(t) = \int_0^L \phi(x)F_e(x, t)dx$; $G_M(t) = \int_0^L \phi(x)F_M(x, t)dx$. Since forcing is non-harmonic over the blade span, the modal GDE is numerically integrated by the state-space method.

$$\{\dot{q}(t-1)\} = GM^{-1}\{GF(t)\} - GM^{-1}GK\{q(t-1)\}; \quad (6a)$$

$$\{\dot{q}(t)\} = \{\dot{q}(t-1)\} + dt\{\ddot{q}(t-1)\}; \quad \{q(t)\} = \{q(t-1)\} + dt\{\dot{q}(t)\} \quad (6b, 6c)$$

For a stable scheme, the time-step is considered from the lowest vortex-shedding time-period, i.e., $dt = 0.01/\max(f_{viv})$. Once the principal coordinates are known, they are multiplied with the modeshapes to generate the total displacement, i.e., $X(x, t) = \sum_{j=1}^6 \phi_j(x)q_j(t)$. The dynamic stresses in flap-wise bending, edge-wise bending, and torsion are: $\sigma_F(x, t) = \sum_{j=1}^2 EI_{zz}(x)\phi_{uj}''(x)q_j(t)$;

$$\sigma_E(x, t) = \sum_{j=1}^2 EI_{yy}(x)\phi_{wj}''(x)q_j(t); \quad \tau_T(x, t) = \sum_{j=1}^2 GJ(x)\phi_{\theta j}'(x)q_j(t) \quad (7)$$

4. Excitation Forces: Computational Fluid Dynamics

The vortex-shedding around 2D aerofoils was investigated in ANSYS FLUENT. The computational setup consisted of a rectangular static domain coupled with a C-shaped moving domain including the aerofoil. A structured quadrilateral mesh was employed for both. The moving mesh was designed to be 1.5 times finer than the background mesh, providing enhanced resolution of near-body flow features and mitigating dynamic mesh movement errors. Inflation layers were applied along the aerofoil and the first cell height selected to maintain $y^+ < 1$, ensuring compatibility with the URANS $k - \omega$ SST turbulence model. Grid independence was verified within the range of 4.8–5.4 million cells, beyond which residuals were $< 1\%$. The model was adopted due to its robustness in capturing adverse pressure gradients and separated flow seen in vortex-shedding. The aerofoil was allowed to oscillate transversely due to instantaneous aerodynamic forces. The mean chord length $\overline{c(x)}$ from **Table 2** was used. The details are in **Table 4**.

Table 4 : Numerical parameters in ANSYS-FLUENT

Description	Magnitude	Description	Magnitude
Turbulence model	$k - \omega$ SST	Overset y^+	0.91
Total Grid Size	5.4 million cells	Motion type	1 DOF translation
Overset Domain Shape	C-shaped	First cell height	2.8×10^{-5} m
Overset upstream, downstream	$2.5c, 4c$	Convergence criterion	Residuals $< 10^{-6}$
Background upstream, downstream	$10c, 40c$	Solution Method	SIMPLE
Inlet boundary condition	Uniform Velocity	Grid Independence Error	$< 1\%$
Outlet boundary condition	Gauge pressure = 0	Blockage ratio	$< 3\%$

5. Results

5.1 Free vibration: Blade Natural frequencies and Modeshapes

Table 5 shows the free coupled flexural-torsional vibration analysis of the blade results in the first six (06) natural frequencies $f = \omega/2\pi$. The fundamental frequencies are close to the ones given in **Table 1**. These are checked for resonance with the vortex-shedding frequencies, which are functions of wind speed and pitching angle. **Figure 5** shows the first two (02) coupled modeshapes $\Phi_j(x)$ from flap-wise (F), edge-wise (E), and torsional (T) vibration, generated from the Rayleigh-Ritz method.

Table 5 : Blade Natural frequencies (Hz)

Mode	Frequency (Hz)
1 st Flap-wise	0.52
1 st Edge-wise	0.74
1 st Torsion	1.28
2 nd Flap-wise	2.85
2 nd Edge-wise	3.20
2 nd Torsion	7.99

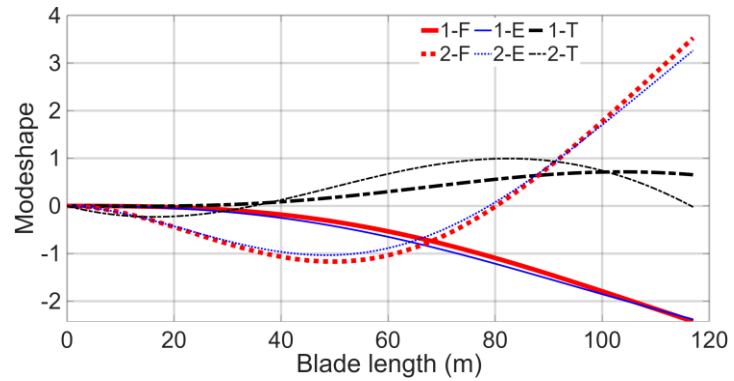


Figure 5 Modeshapes : 1F = 1st Flap-wise, 1E = 1st Edge-wise, 1T = 1st Torsion, 2F = 2nd Flap-wise, 2E = 2nd Edge-wise, 2T = 2nd Torsion.

5.2 Tuning factor check: predictive analysis

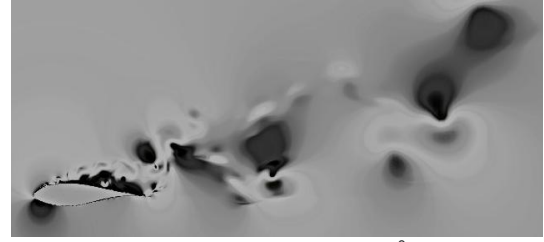
Table 6 shows the tuning factors $\Lambda = f_{viv}/f$ of the vortex-shedding at the rated wind speed of $U = 10.59 \text{ m/s}$, and various pitching angles; against the first modeshapes of flap-wise, edge-wise, and torsional vibration, respectively. Since f_{viv} varies along the span, a mean vortex-shedding frequency for each sub-span is considered for the resonance check. The Λ is checked for $U = 3, 7, 10.59, 15, 25 \text{ m/s}$. At each wind speed, five (05) different pitch angles $\delta = 105^\circ, 95^\circ, 85^\circ, 75^\circ, 60^\circ$ are considered. The critical zone of $0.8 < \Lambda < 1.2$ has been highlighted, showing that the resonant vortex-shedding can be at different sub-spans. The location of the resonant vortex-shedding depends on U, δ .

Table 6 : Tuning factor for $U = 10.59 \text{ m/s}$, at pitching angles $\delta = 105^\circ, 95^\circ, 85^\circ, 75^\circ, 60^\circ$.

	$\delta = 105^\circ$			$\delta = 95^\circ$			$\delta = 85^\circ$			$\delta = 75^\circ$			$\delta = 60^\circ$		
	1F	1E	1T	1F	1E	1T	1F	1E	1T	1F	1E	1T	1F	1E	1T
Aerofoil	0.52	0.74	1.28	0.52	0.74	1.28	0.52	0.74	1.28	0.52	0.74	1.28	0.52	0.74	1.28
Circular	0.76	0.53	0.31	0.76	0.53	0.31	0.76	0.53	0.31	0.76	0.53	0.31	0.76	0.53	0.31
SNL-FFA-W3-500	0.71	0.50	0.29	1.07	0.75	0.43	1.42	1.00	0.58	1.42	1.00	0.58	0.71	0.50	0.29
FFA-W3-360	0.76	0.53	0.31	1.13	0.80	0.46	1.51	1.06	0.61	1.51	1.06	0.61	0.76	0.53	0.31
FFA-W3-330blend	1.28	0.90	0.52	1.28	0.90	0.52	1.71	1.20	0.69	1.71	1.20	0.69	0.85	0.60	0.35
FFA-W3-301	1.45	1.02	0.59	1.45	1.02	0.59	1.93	1.36	0.79	1.45	1.02	0.59	0.97	0.68	0.39
FFA-W3-270blend	1.65	1.16	0.67	1.65	1.16	0.67	2.20	1.54	0.89	1.65	1.16	0.67	1.10	0.77	0.45
FFA-W3-241	1.91	1.34	0.77	2.54	1.79	1.03	2.54	1.79	1.03	1.91	1.34	0.77	1.27	0.89	0.52
FFA-W3-211	2.76	1.94	1.12	3.68	2.58	1.49	3.68	2.58	1.49	2.76	1.94	1.12	1.84	1.29	0.75

5.3 CFD of 2D aerofoils: Coefficients of Drag, Lift, Moment with corrections.

The CFD simulations using dynamic mesh were used to generate the coefficients of aerodynamic lift, drag, and moment C_L, C_D, C_M ; at various AoA and all 7 aerofoils, for $Re = 10^6$. The pitch angle δ is varied over the feathering range $60^\circ \leq \delta \leq 105^\circ$, with a closer investigation around aerodynamic braking. Stalling pitch is avoided because it does not lead to alternate vortex-shedding. **Figure 6** shows the vortex patterns around aerofoil (FFA-W3-211) at two distinct AoA. Larger AoA reduce the streamlined profile of the aerofoil with respect to the flow, making it behave like a bluff body.


 Figure 6(a) $\alpha = 10^\circ$

 Figure 6(b) $\alpha = 50^\circ$

The spanwise distribution of VIV indicates that excitation is minimal at the root and tip, while the inboard-to-mid sections dominate, with ΔC_L rapidly increasing from the root and peaking in the mid-to-outer span (due to optimal geometric and aeroelastic conditions for coherent shedding), followed by a reduction near the tip due to 3D flow relief. Pitch angle δ governs regime transition, with a critical VIV-active zone between 60° – 75° characterized by sustained high ΔC_L (lock-in), while higher δ (95° – 105°) suppress oscillatory lift and induce aerodynamic damping despite elevated drag. Wind speed exhibits a non-monotonic but robust influence, where intermediate velocities (~ 7 – 15 m/s) consistently sustain VIV due to resonance, whereas low speeds under-energize shedding, and higher speeds shift the flow toward intensified turbulence with increased drag but not proportionally higher lift oscillations. Collectively, the results confirm that VIV is confined to a coupled instability envelope defined by mid-span amplification, stall-induced flow separation, and resonance-prone wind regimes.

6. Root Stresses with Machine Learning.

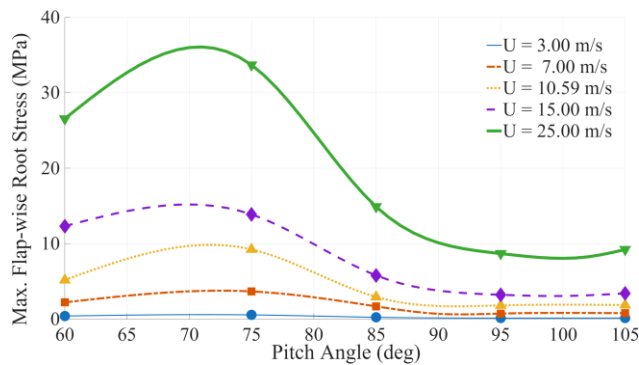


Figure 7(a) Flap-wise Root Bending Stress.

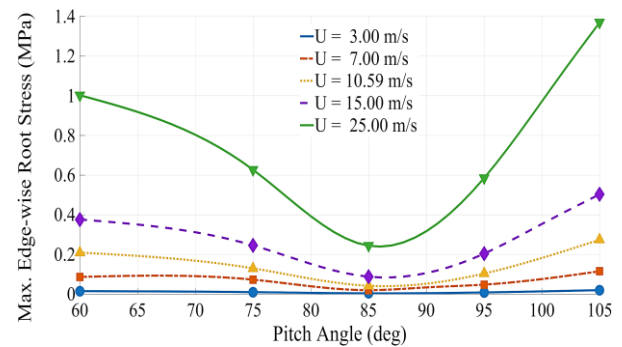


Figure 7(b) Edge-wise Root Bending Stress.

Figure 7 shows the maximum flap-wise and edge-wise root bending stresses; as functions of wind speed U and pitch angles δ . The torsional stress maxima decrease monotonically with increasing δ . This data was initially evaluated for a discrete set of 25 operating configurations (Section 5.2). The markers on each graph show the dynamic response from the modal superposition approach (Section 3.2). To extend the discrete simulation results across a continuous parameter space, a Gaussian Process Regressor (GPR) was implemented. GPR was selected for its strong suitability for sparse, high-fidelity datasets, and its probabilistic prediction capability. It is a non-parametric, kernel-based method that can effectively capture highly nonlinear relationships between multiple inputs (U , δ , C_D , C_L , C_M , ΔC_D , ΔC_L , ΔC_M) and outputs (root stresses). The advantages of GPR applicable to this type of data set are highlighted by Forrester et al (2008). The model was trained using 25 high-fidelity structural data points, augmented with a dense aerodynamic coefficient database spanning U and δ , enabling physically informed learning of stress behaviour. A Matérn ($\nu = 3/2$) kernel was employed, with hyperparameters optimized via maximum likelihood estimation. Model performance was evaluated using an 80–20 train–test split and 5-fold cross-validation, yielding a consistent RMSE of $\sim 0.8\%$. The trained GPR model was subsequently used to predict root stresses at intermediate pitch angles not explicitly simulated, enabling continuous stress mapping across the operating envelope.

7. Discussion and Conclusion

The blade root VIV loading is dominated by the flap-wise stress, which significantly outweighs edge-wise and torsion components due to low flap-wise stiffness. At $U = 25 \text{ m/s}$, flap-wise stress peaks to 36.06 MPa (at $\delta = 71^\circ$), while edge-wise stress remains negligible ($< 1.4 \text{ MPa}$). Load-to-pitch sensitivity is highly non-linear and wind-speed dependent: while stresses are almost constant for pitching at $U = 3 - 7 \text{ m/s}$, they increase rapidly at $U = 15 - 25 \text{ m/s}$ for non-optimal pitch angles. Crucially, mid-to-outer blade sections are the primary sources of VIV excitation. Consequently, VIV-induced fatigue is not a function of highest U alone, but of a narrow pitch–wind–span envelope that maximizes oscillatory forcing. VIV risk is highest when shedding frequencies align with natural frequencies, typically at $\delta = 60^\circ - 75^\circ$. At 25 m/s, the standard $\delta = 90^\circ$ feathering position yields a flap-wise stress of 10.11 MPa. However, over-feathering to $\delta = 100^\circ$ reduces this to a minimum of 8.05 MPa, representing a 20% load reduction compared to standard feathering. This confirms that $\delta = 100^\circ$ as the dynamically resilient setpoint. This transition toward feathered configurations ($\delta \geq 95^\circ$) triggers a sharp drop in the flap-wise stress by disrupting the coherent vortex-shedding that drives oscillatory lift. A robust, multi-layered mitigation strategy is therefore proposed to ensure structural longevity by avoiding the $60^\circ - 75^\circ$ instability envelope. The recommended pitching approach would be to utilize a standard 90° feather at wind speeds below 10 m/s, and transition to $95^\circ - 105^\circ$ stability bucket as wind speeds increase. For further improved recommendations, future work would include full 3D blade CFD, two-way coupled FSI, fatigue damage estimates, rotor-tower coupled response, and application of physics-based ML approaches.

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